

To complete this **optional** activity, you must follow the step-by-step instructions and receive at least 75%, or 3 out of 4 points, on the questions that follow. Once you have answered the questions, review the feedback statements. You can learn more about the graded and practice items in the [course overview](#).



Activity Overview

In this activity, you'll use the Wazuh platform to perform basic searches on a data set and answer a series of questions. The preceding reading, "Follow-along guide for Wazuh setup," includes instructions to prepare for this activity. You should have already completed the following steps:

- Downloaded and configured the Wazuh virtual machine
- Ingested the tutorialdata.zip file into your Wazuh instance

Please note that this activity is optional and will not affect your completion of the course.

So far, you've learned that SIEM tools, such as Splunk, and Wazuh, are an important part of a security analyst's toolbox because they provide a platform for storing, analyzing, and reporting on data from different sources. You also explored basic searches using a querying language that included the use of pipes and wildcards.

Creating effective searches is an important skill because it enables you to quickly and accurately find the information you are looking for within a large amount of data. Quick and accurate searching is especially useful during incident response, because you might need to swiftly identify and address a security incident. Effective search techniques also help you efficiently identify patterns, trends, and anomalies within data.

Scenario

Review the following scenario. Then complete the step-by-step instructions.

You are a security analyst working at the e-commerce store Buttercup Games. You've been tasked with identifying whether there are any possible security issues with the mail server. To do so, you must explore any failed SSH logins for the root account.

Note: Use the incident handler's journal you started in [a previous activity](#) to take notes during the activity and keep track of your findings.

Step-By-Step Instructions

Follow the instructions and answer the following questions to complete the activity.

▼ Step 1: Access supporting material

The following supporting materials will help you complete this activity. The data contains log and event information from Buttercup Games' mail servers and web accounts. This includes information like access and authentication logs, email logs, and more.



To download this data, click the link then click the download icon.

Link to supporting materials: [tutorialdata.zip](#) file

OR

If you don't have a Google account, you can download the supporting materials directly from the following attachment.



tutorialdata
ZIP File

^ Step 2: Access the Wazuh Dashboard

To begin, ensure your Wazuh virtual machine is running and you have access to the dashboard. You can find your VM's IP address by running `ip a` in its command line. Open your web browser and navigate to `https://<your_VM_IP_address>`.



^ Step 3: Verify and Locate the Data

The data for this activity should have been ingested and should be searchable. In the Wazuh dashboard, navigate to the **Discover** option under the **Explore** option to access the full-featured search bar.

^ Step 4: Perform a basic search

Now that you've located the correct search bar, perform your first query to confirm that the data has been ingested and is searchable. Follow these steps to perform a query:

1. In the time range dropdown at the top right, select **Absolute**. In the calendar, choose an old start date (like January 1, 2000) and click **Update**.
2. In the search bar, enter your search query: `*`.
3. Click the search button (magnifying glass). Your search should retrieve over 100,000 events.

^ Step 5: Evaluate the fields

When Wazuh indexes data, it attaches fields to each event. These fields become part of the searchable event data. This helps



security analysts easily search for and find the specific data they need. Now that you've run your first query, examine the search results and the fields.

For each event, the fields are host, source, and sourcetype. Under the Available fields on the left, you can examine these same fields.

Examine the field values by clicking on the field in the Available fields list. You should observe the following:

- **host.keyword:** The host field specifies the name of the network host from which the event originated. In this search, there are five hosts:
 - mailsv - Buttercup Games' mail server. Examine events generated from this host.
 - www1 - This is one of Buttercup Games' web applications.
 - www2 - This is one of Buttercup Games' web applications.
 - www3 - This is one of Buttercup Games' web applications.
 - vendor_sales - Information about Buttercup Games' retail sales.
- **log.file.path.keyword:** This field indicates the file name from which the event originates. You should identify eight different files. Notice /mailsv/secure.log, which is a log file that contains information related to authentication and authorization attempts on the mail server.



^ Step 6: Narrow your search

Because you've been tasked with exploring any failed SSH logins for the root account on the mail server, you'll need to narrow the search results for events from the mail server.

In the search bar, update your query to filter for events from the mailsv host: `host.keyword: mailsv`

The search results should narrow to over 9,000 events that are generated by the mail server.

^ Step 7: Search for a failed login for root

Now that you've narrowed your search results to events generated by the mail server, continue to narrow the search to locate any failed SSH logins for the root account.

In the search bar, enter the following query: `host.keyword: mailsv AND (fail* OR failed) AND root`

This search expands on the search from the previous task and searches for the keywords fail (using a wildcard to find failure, failed, etc.) and root.

^ Step 8: Evaluate the search results

Your search from the previous task should have retrieved search results for over 300 events. Navigate to other pages of the search results to observe the events not listed on the first page of results.

Pro tip: The Wazuh dashboard highlights search terms in search results to make it easier to identify where the search terms appear in the data.

Step 9: Answer questions about the search results

1. How many events are contained in the buttercup-games* index across all time?

1 / 1 point

- ☐ 10,000
- ☐ 10 - 99
- ☐ 100 - 1,000
- ☒ Over 100,000

✓ Correct

When you enter the search query * in the Wazuh dashboard's Discover tab with an "Absolute" time range set to an early date, the total number of events returned is 109,864, which is over 100,000.

2. Which field identifies the name of a network device or system from which an event originates?

1 / 1 point

- ☒ host
- ☐ source
- ☐ sourcetype
- ☐ index

✓ Correct

The host.keyword field in the Wazuh dashboard identifies the name of the host, such as a network device or other system, from which an event originates.

3. Which of the following hosts used by Buttercup Games contains log information relevant to financial transactions?

1 / 1 point

- ☐ www3
- ☐ www1
- ☐ www2
- ☒ vendor_sales

✓ Correct

By examining the values associated with the host.keyword field, you can determine that the vendor_sales host provides information about Buttercup Games' retail sales, making it relevant to financial transactions.

4. How many failed SSH logins are there for the root account on the mail server?

1 / 1 point

- ☐ One
- ☒ More than 100
- ☐ 100
- ☐ None

✓ Correct

When you run the search query host.keyword: mailsrv AND (fail* OR failed) AND root, the Wazuh dashboard shows a total of 376 events, which is more than 100.

Key takeaways In this activity, you used the Wazuh dashboard to perform a search and investigation. Using Wazuh, you were able to:

- Ingest sample log data
- Search through indexed data

4. How many failed SSH logins are there for the root account on the main server?

4 / 4 points

- ☐ One
- ☒ More than 100
- ☐ 100
- ☐ None

✔ Correct

When you run the search query `host.keyword: mailsrv AND (fail* OR failed) AND root`, the Wazuh dashboard shows a total of 376 events, which is more than 100.

Key takeaways In this activity, you used the Wazuh dashboard to perform a search and investigation. Using Wazuh, you were able to:

- Ingest sample log data
- Search through indexed data
- Evaluate search results
- Identify different data sources
- Locate failed SSH login(s) for the root account